Introduction to Probability and Statistics

Solutions to Exercises Lecture 2

Israel Leyva-Mayorga

Exercise 1: Repeat the last exercise of the lecture with different parameters: There's a bag with 3 balls. Each ball is either orange or green. The number of green balls is $\theta \in \{0, 1, 2, 3\}$. Estimate θ by grabbing and putting back a ball 5 times for the cases:

a) 4 out of the 5 of the balls are green. Calculate the likelihood for each possible value of $\theta \in \mathbb{N}$ (i.e., θ must be an integer).

b) 4 out of the 5 of the balls are green. Calculate the likelihood and maximize the likelihood function for $\theta \in \mathbb{R}$ and then choose the best valid integer.

c) 3 out of the 5 of the balls are green. Calculate the likelihood for each possible value of $\theta \in \mathbb{N}$ (i.e., θ must be an integer).

d) 3 out of the 5 of the balls are green. Calculate the likelihood and maximize the likelihood function for $\theta \in \mathbb{R}$ and then choose the best valid integer.

Solution:

a) When we observe the five outcomes before doing the estimation we use the joint pmf for a given value of parameter θ

$$P_{X_1,X_2,X_3,X_4,X_5}(x_1, x_2, x_3, x_4, x_5; \theta) = \prod_{i=1}^{5} P(X_i = x_i; \theta) \tag{1}$$

For simplicity, I'll assume that the first four outcomes are the same as in the example from the lecture and only the last one will be added. In this case, the last observed ball is green. Then, we calculate this joint probability for the observed outcomes and each of the possible answers $\theta =\in \{0, 1, 2, 3\}$.

- *For* $\theta = 0$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1, 0, 1, 1, 1; \theta = 0) = 0 \times 1 \times 0 \times 0 \times 0 = 0 \tag{2}$$

- *For* $\theta = 1$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1, 0, 1, 1, 1; \theta = 1) = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = 0.00823 \tag{3}$$

- *For* $\theta = 2$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1,0,1,1,1;\theta=2) = \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = 0.06584 \tag{4}$$

- *For* $\theta = 3$*:* we obtain 0.

So, the best guess, considering the likelihood of the possible values for θ is $\hat{\theta}_5 = 2$.

b) We use the same log-likelihood function as in the lecture, which is

$$\ell_n(\theta) = X_n\left(\log(\theta) - \log(3)\right) - (n - X_n)\log\left(1 - \frac{1}{3}\right) \tag{5}$$

You can see this plotted in Fig. 1. We also know that the Maximum Likelihood Estimator (MLE) is

$$\hat{\theta}_n = 3X_n/n, \tag{6}$$

which for our case is $\hat{\theta}_5 = 12/5 = 2.4$. So, our valid solution is $\hat{\theta}'_5 = 2$.

c) We repeat the process with the last outcome being an orange ball, so $X_5 = 0$

- *For* $\theta = 0$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1,0,1,1,0;\theta=0) = 0 \times 1 \times 0 \times 0 \times 0 = 0 \tag{7}$$

- *For* $\theta = 1$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1,0,1,1,0;\theta=1) = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = 0.01646 \tag{8}$$

- *For* $\theta = 2$*:*

$$P_{X_1,X_2,X_3,X_4,X_5}(1,0,1,1,0;\theta=2) = \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = 0.03292 \tag{9}$$

- *For* $\theta = 3$*:* we obtain 0.

Again, the best guess, considering the likelihood of the possible values for θ is $\hat{\theta}_5 = 2$. However, the decision is closer now between $\theta = 1$ and $\theta = 2$ than in a).

d) From the same MLE as in b), we calculate $\hat{\theta}_5 = 9/5 = 1.8$. So, our valid solution is again $\hat{\theta}'_5 = 2$. However, the log-likelihood function for this case, shown in Fig. 1, is less steep than for the case where 4 green balls were drawn.

Exercise 2: The room temperature during a PhD course was recorded using a Raspberry Pi and a temperature sensor. The collected values are in the file temperature.csv (the temperature is the second column).

a) Calculate the sample mean $\overline{X}_n$ and variance S_n^2 of the temperature with all the $n = 253$ measurements

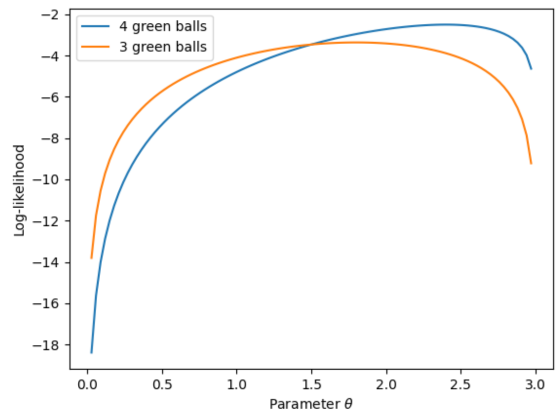


Fig. 1. Log-likelihood function when observing 4 and 3 green balls in 5 experiments.

b) Assume that $\overline{X}_n$ is the true temperature μ and that S_n^2 is the real variance of the Gaussian noise that affects each measurement of the sensor, denoted by σ^2. Consequently, assume that the temperature measurements have a distribution $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Plot the likelihood or log-likelihood function for the first 10 and with the first 100 measurements

c) Calculate the MLE for μ with the first 10 and with the first 100 measurements under the previous assumption.

Solution:

a) Using all the $n = 253$ samples, these give $\overline{X}_n = 26.720$ and variance $S_n^2 = 0.112$

b) Given that the probability density function (PDF) of the Gaussian random variable (RV) is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{(x-\mu)^2}{2\sigma^2}},$$

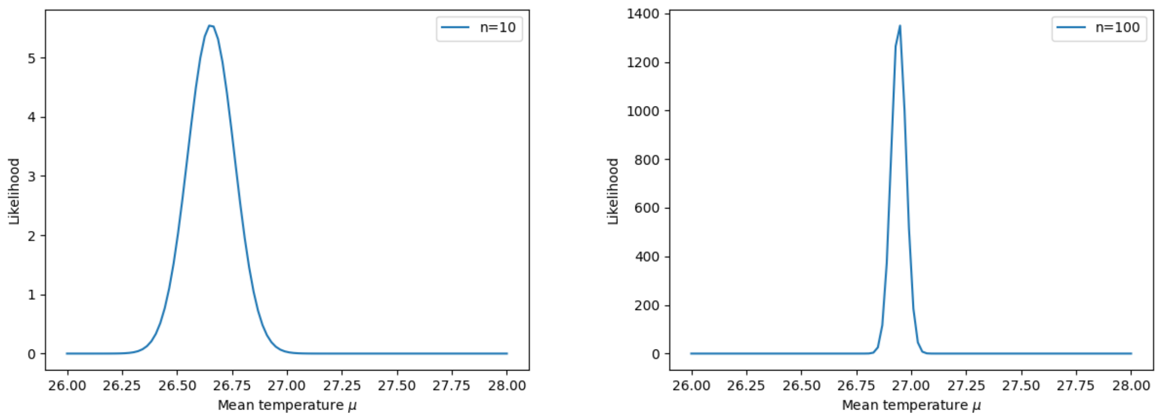
we calculate the likelihood function as

$$\mathcal{L}_n(\mu) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X_i-\mu)^2}{2\sigma^2}} = \frac{e^{\left(\sum_{i=1}^{n} \frac{-(X_i-\mu)^2}{2\sigma^2}\right)}}{\left(\sigma\sqrt{2\pi}\right)^n}$$

and the log-likelihood as

$$\ell_n(\mu) = \log\left(\mathcal{L}_n(\mu)\right) = -n(\sigma + \sqrt{2\pi}) - \sum_{i=1}^{n} \frac{(X_i-\mu)^2}{2\sigma^2}.$$

By plotting the likelihood and log-likelihood functions with the first $n = \{10, 100\}$ measurements, we get the plots shown in Fig. 2 and Fig. 3.



(a) (b)

Fig. 2. Likelihood for the temperature with (a) $n = 10$ and (b) $n = 100$, given that it is a Gaussian RV.

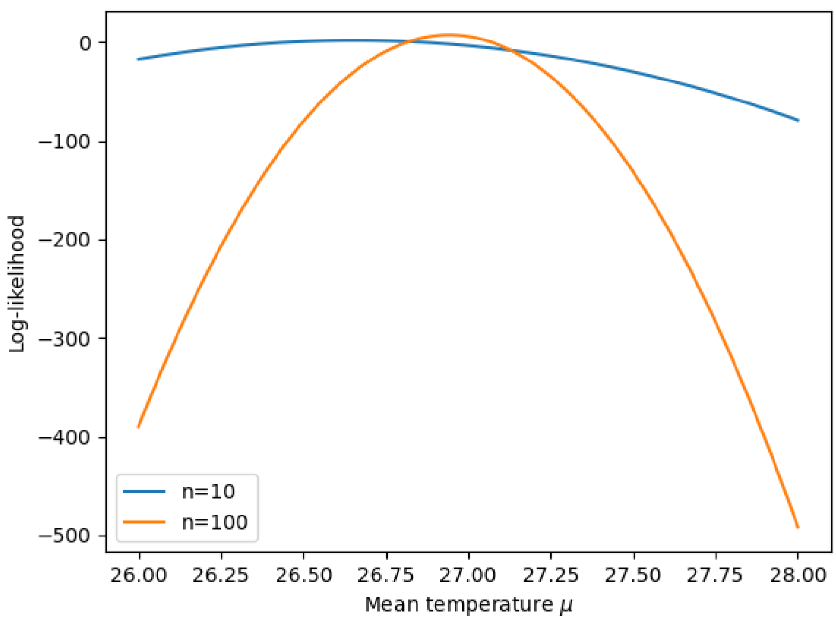


Fig. 3. Log-likelihood for the temperature, given that it is a Gaussian RV.

c) The MLE is obtained by setting $X = \sum_{i=1}^{n} X_i$ and finding the arg max of

$$\frac{\partial \ell_n(\mu)}{\partial \mu} = \frac{2X - 2n\mu}{2\sigma^2} = 0, \longrightarrow \hat{\mu}_n = \frac{X}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}_n$$

Therefore, the MLE for $n = 10$ gives $\hat{\mu}_{10} = 26.655$ and for $n = 100$ gives $\hat{\mu}_{100} = 26.943$.